1) Let us consider an orbit Velocity frame $\mathcal{V} : \{\hat{i}_n, \hat{i}_v, \hat{i}_h\}$ which differs from the Orbit frame, $\mathcal{O} = \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$ through a single-axis rotation about $\hat{i}_h$ as illustrated in Figure 1. Your task is to derive an expression for the direction cosine matrix, $C_{\mathcal{VO}}$ as a function of eccentricity of the orbit, e and true anomaly, f. Furthermore, calculate the angular velocity and angular acceleration of the velocity frame with respect to the Perifocal frame as a function of eccentricity of the orbit, e and true anomaly, f.

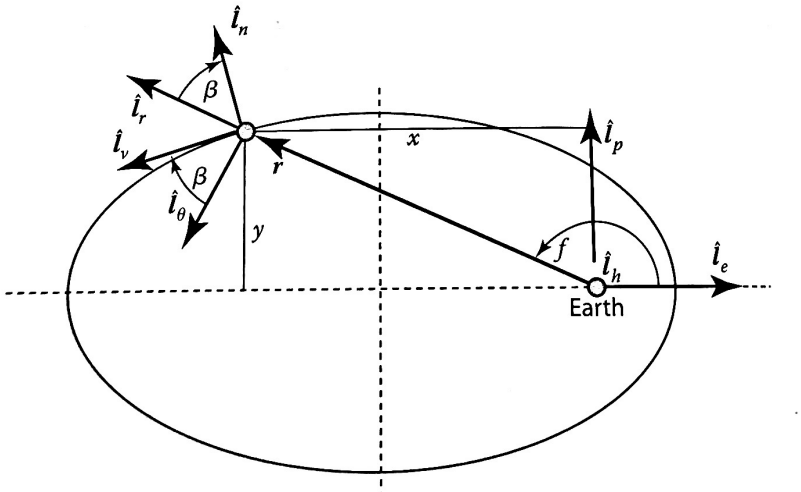


Fig. 1. A Schematic of Velocity Frame

2) A space probe is in an elliptical orbit around the sun. The perihelion distance is $0.5AU$ and the aphelion is $2.5AU$. How much time during each orbit is the probe closer than $1AU$ to the sun?

3) If r_a and r_p are the apoapsis and perapsis of a satellite orbit around the Earth, then prove the following:
   - $a = \frac{1}{2}(r_p + r_a)$
   - $b = \sqrt{r_p r_a}$
   - $p = \frac{2r_p r_a}{r_p + r_a + a}$

4) Let us consider the location of a satellite at two time instants in the same orbit around the Earth:

$$\bar{r}_1 = 14450.6\hat{n}_1 - 1529.9\hat{n}_2 - 6524.0\hat{n}_3$$

$$\bar{r}_2 = -6199.5\hat{n}_1 + 14699.2\hat{n}_2 + 8531.9\hat{n}_3$$

   The semi-latus rectum of the orbit is $p = 2.88R_e$ with $R_e = 6378.137$ km being the radius of the Earth. Your task is to compute $a, e, i, \Omega, \omega, f_1, f_2$. Also, comment on whether the satellite is ascending or descending at the two locations.

5) Problems from Text Book: 2.16, 2.17, 2.19, 2.20, 2.21, 2.22, 2.28, 2.32, 2.42